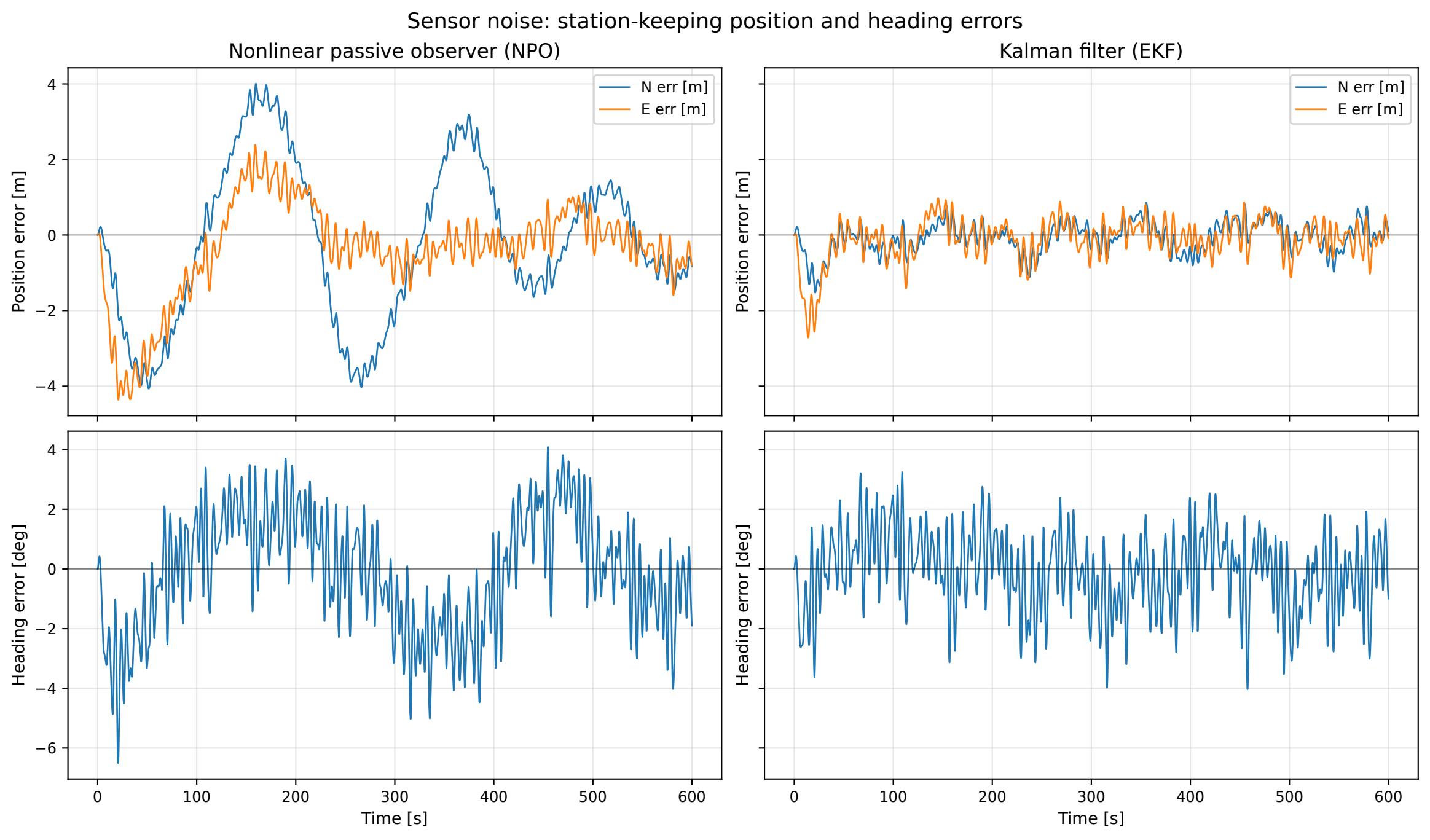

Sensor noise: station-keeping position and heading errors
Nonlinear passive observer (NPO)
Kalman filter (EKF)
N err [m]
E err [m]
Position error [m]
Heading error [deg]
Time [s]
4
2
0
−2
−4
−6
0
100
200
300
400
500
600